Note Title 3/3/2018

Unit 2.4 Continuity (p. 115 – 126)

NB! Definition: (p. 115) (NB: Read text book)

For f to be continuous at $x=a$, we need 3 steps (p. 115)

1. $f(a)$ must exist
2. $\lim_{x\to a} f(x)$ must exist, that means: $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x)$
3. $\lim_{x\to a} f(x)$ must be equal to $f(a)$

If one of these fails, f is discontinuous at $x=a$.

Examples:

1. Show that $f(x)=x^2-1$ is continuous at $x=2$.
2. $f(x)=\begin{cases}\frac{x^2-9}{x-3} & \text{if } x\neq 3\\ 6 & \text{if } x=3.\end{cases}$ Is f continuous in $x=5$?
3. $f(x)=\begin{cases}\frac{3x}{x^2+x} & \text{if } x>0\\ 3 & \text{if } x=0\\ \frac{\sin(3x)}{x} & \text{if } x<0\end{cases}$ Is f continuous at $x=0$?
4. $f(x)=\begin{cases}x^2 & \text{if } -1\leq x\leq 2\\ ax+b & \text{if } x<-1 \text{ or } x>2.\end{cases}$
   Determine a and b such that f is continuous on $\mathbb{R}$.
5. $f(x)=\frac{x^2-4}{x-2}$. Is f continuous at $x=2$?
6. Find the value of a if f is continuous at $x=5$.
   $f(x)=\begin{cases}ax^2+2x & \text{if } x\leq 5\\ x^2-ax & \text{if } x>5.\end{cases}$

If f is not continuous at a, we say that f is discontinuous at $x=a$.

Removable discontinuity, $f(x)=\frac{x^2-x-2}{x-2}$ as example

Infinite discontinuity, e.g. $f(x)=\begin{cases}\frac{1}{(x-1)^2} & \text{if } x\neq 1\\ 0 & \text{if } x=1.\end{cases}$

Jump discontinuity, e.g. $f(x)=\begin{cases}1 & \text{if } x\geq 0\\ -1 & \text{if } x<0.\end{cases}$

Definition p. 117: Continuity from the right / left.
Definition p. 117: Continuity on an interval.

Know Theorem 4 (p. 118)

Do example 6 on p. 120

NB

Theorem 8: (p. 120)

If $\lim_{x\to a} g(x) = b$ and f is continuous on b,

Then: $\lim_{x\to a} f(g(x)) = f(\lim_{x\to a} g(x)) = f(b)$.

Do example 8, p.121.

NB

Theorem 9 (p.121)

Continuity of a composite function

Example:

Where is $h(x) = \sin(e^x)$ continuous?

Suppose $h(x) = f(g(x))$ where $f(x) = \sin(x)$ and $g(x) = e^x$.

Now: $g(x)$ is continuous on $\mathbb{R}$, since it is the exponential function and f is also continuous on $\mathbb{R}$, i.e. $f \circ g$ is continuous on $\mathbb{R}$, by Theorem 9.